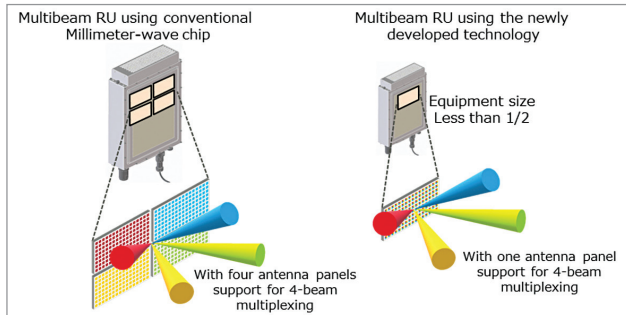




a standard RU. This approach uses fewer millimetre-wave chips, resulting in a 30% reduction in energy usage per RU. The initiative focused on crafting systems that bolster 5G

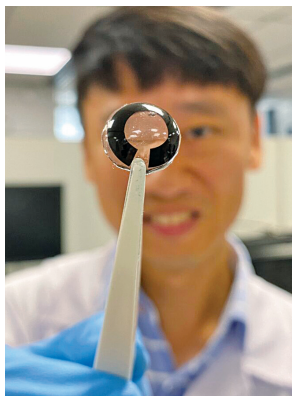


Comparison image of RU using a conventional millimeter-wave chip and RU applying this technology (Credit: www.fujitsu.com)

support. The team's efforts resulted in a breakthrough technology capable of multiplexing multiple beams on a single chip via millimetre-wave beamforming. This means an RU can now offer four times the speed and greater capacity than its conventional counterpart of equal size, all while bringing about significant energy and cost savings.

Contact lenses with adaptive batteries fuelled by saline solution

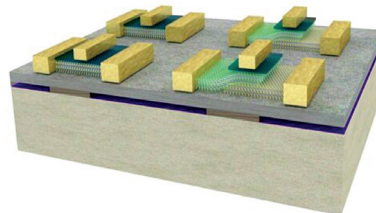
Scientists from Nanyang Technological University (NTU) Singapore have developed a flexible battery as thin as a human cornea, which stores electricity when immersed in saline solution. These smart lenses not only correct vision but also monitor the wearer's health metrics continuously through integrated sensors and data transmission systems. They can track glucose levels in tears and monitor intraocular pressure, offering a non-invasive solution for individuals with diabetes or glaucoma. Additionally, they can alert users about potential health issues, remind them about medication, or provide augmented reality overlays for daily assistance, revolutionising both tech and healthcare sectors. The battery, made of biocompatible materials, eschews wires and toxic heavy metals found in conventional batteries or wireless charging systems. Its glucose-based coating interacts with the saline solution's sodium and chloride ions, using its contained water as a circuit to generate electricity.



Scientists have developed an ultra-thin battery powered by saline for smart contact lenses (Credit: NTU Singapore)

Advancing neuromorphic computing by fusion of analogue and digital approaches

Emerging from a partnership between the Nanoelectronics Device Laboratory (Nanolab) and the Microsystems Laboratory, the innovation focuses on merging materials



that introduce brain-like characteristics and state-of-the-art electronic switches, specifically the notable negative capacitance tunnel field-effect transistor (TFET). The TFET demonstrates the potential to integrate synaptic neuron functions using the same technology. It harmoniously combines 2D semiconductors and ferroelectric materials into a unified electronic system. 2D semiconductors facilitate ultra-efficient digital processors, and incorporating ferroelectric materials allows for continual processing and memory storage. This combination leverages the strengths of both digital and analogue functionalities, making the metaphorical light switch more energy-efficient with increased brightness. The fusion hints at the emergence of electronic devices that could operate parallel to the human brain, combining rapid computation with a cognitive approach to processing information, potentially excelling in tasks like pattern recognition and sensory data processing.

By seamlessly integrating ultra-thin, two-dimensional semiconductors with ferroelectric materials, the research, published in *Nature Electronics*, unveils a novel way to improve energy efficiency and add new functionalities in computing. The new configuration merges traditional digital logic with brain-like analogue operations (Credit: EPFL)

By seamlessly integrating ultra-thin, two-dimensional semiconductors with ferroelectric materials, the research, published in *Nature Electronics*, unveils a novel way to improve energy efficiency and add new functionalities in computing. The new configuration merges traditional digital logic with brain-like analogue operations (Credit: EPFL)

Brain implants transform thoughts into speech

Scientists from Radboud University and the University Medical Centre Utrecht (UMC Utrecht) have successfully showcased the capability to convert brain signals into recognisable speech by utilising a brain-machine interface paired with an artificial intelligence (AI) system. Researchers focused on non-paralysed individuals with temporary brain implants, asking them to verbalise given words while monitoring brain signals. Analysing this data, the team found a direct correlation between brain activity and speech, which they incorporated into AI models to convert thoughts into speech. In trials, the system successfully reconstructed speech sentences from brain signals, demonstrating the potential to help individuals with speech disabilities or serve as a covert communication tool in high-stress environments.